

15. (2.25 points)

Given the following information for a retrospectively rated book of business:

- Basic Premium is \$25 million.
- Standard Premium is \$100 million.
- Tax Multiplier is 1.03.
- Loss Conversion Factor is 1.18.
- Expected Loss Ratio is 0.8.

Retro Adjustment Period	Incremental Loss Capping Ratio	Age-to-Ultimate Loss Development Factor
1st	0.95	2.00
2nd	0.75	1.08
3rd	0.55	1.01

a. (1.5 points)

Calculate the cumulative premium development to loss development (PDL) ratio at the 1st retro adjustment.

b. (0.25 point)

Briefly describe a different approach for estimating PDL ratios for retrospectively rated policies.

c. (0.5 point)

Briefly describe an advantage and a disadvantage of using the method in part a. above over the method described in part b. above.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 15											
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE(S): A14										
SAMPLE ANSWERS											
Part a: 1.5 points											
<u>Sample 1</u>											
$PDLD1 = \left(\frac{25}{100}\right) * \frac{1.03}{.8 * .5} + (.95 * 1.18 * 1.03) = 1.798$ $PDLD2 = .75 * 1.18 * 1.03 = .912$ $PDLD3 = .55 * 1.18 * 1.03 = .668$ $CPDLD1 = \frac{(1.798 * .5 + .912 * .426 + .668 * .064)}{.5 + .426 + .064 + .01} = 1.330$	<table border="1"> <thead> <tr> <th>Adj #</th><th>Emergence</th></tr> </thead> <tbody> <tr> <td>1</td><td>.5</td></tr> <tr> <td>2</td><td>.426</td></tr> <tr> <td>3</td><td>.064</td></tr> <tr> <td>4</td><td>.01</td></tr> </tbody> </table>	Adj #	Emergence	1	.5	2	.426	3	.064	4	.01
Adj #	Emergence										
1	.5										
2	.426										
3	.064										
4	.01										
<u>Sample 2</u>											
$PDLD1 = \left(\frac{25}{100}\right) * \frac{1.03}{.8 * .5} + (.95 * 1.18 * 1.03) = 1.798$ $PDLD2 = .75 * 1.18 * 1.03 = .912$ $PDLD3 = .55 * 1.18 * 1.03 = .668$ $CPDLD1 = \frac{(1.798 * .5 + .912 * .426 + .668 * .064)}{.5 + .426 + .064} = 1.34$	<table border="1"> <thead> <tr> <th>Adj #</th><th>Emergence</th></tr> </thead> <tbody> <tr> <td>1</td><td>.5</td></tr> <tr> <td>2</td><td>.426</td></tr> <tr> <td>3</td><td>.064</td></tr> <tr> <td>4</td><td>.01</td></tr> </tbody> </table>	Adj #	Emergence	1	.5	2	.426	3	.064	4	.01
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1	.5										
2	.426										
3	.064										
4	.01										
<u>Sample 3</u>											
<i>(Same Calculations for PDLD1, PDLD2, PDLD3 as above)</i>											
$CPDLD3 = \frac{(.668 * .064)}{.064 + .01} = .578$ $CPDLD2 = \frac{.912 * .426 + .578 * (.064 + .01)}{.426 + .064 + .01} = .863$ $CPDLD1 = 1.798 + .863 * .5 = 1.33$											
Part b: 0.25 point											
<u>Sample Responses</u>											
• Use Booked Premium Triangle and Loss Development Triangle to generate PDLDs • Select PDLD ratios based on empirical analysis • Use Fitzgibbon method $P=a+bx$ • Fit a linear model comparing UltLosses(X) to Ult Premium(Y) -> $Y=aX+b$											
Part c: 0.5 point											
<u>Sample Responses</u>											
If part b was the empirical method: Advantages of formula method over empirical • Responds to changing retro parameters that are being sold • More stable than empirical approach • Easier to explain to underwriters • Hard to estimate retro parameters if there is a mix of different retro polices • Can calculate PDLD faster, do not have to wait for premium lag											

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Disadvantages of formula to empirical

- Difficult to estimate the retro parameters (Tax multiplier by state etc.)
- May not reflect trends in the data in a timely manner
- Ignores the actual relationship between the premium emerged and the reported losses
- Using average factors could lead to bias in results
- Portfolio level factors may not be appropriate for individual accounts

If part b was Fitzgibbon/Linear approximation:

Advantage of formula over Fitzgibbon:

- Recognizes that responsiveness of premium development to loss development is not linear
- Formula recognizes declining premium responsiveness in later adjustments

Disadvantage of formula to Fitzgibbon

- Difficult to measure parameters
- More calculation intensive
- More time consuming

EXAMINER'S REPORT

The candidates need to be able to calculate the cumulative premium development to loss development ratio using the formula method laid out in the Teng and Perkins paper. They also need to be aware of a separate method to estimate the PDLD.

Part a

Candidates were expected to know how to calculate the PDLD at the first adjustment. They were expected to know how to use the incremental percent reported to get to the cumulative PDLD (CPDLD) for the first adjustment.

The question required candidates to calculate the cumulative premium development to loss development ratio. Some candidates incorrectly interpreted "(PDLD)" in the question to mean that only the PDLD at the first adjustment was required, and did not calculate the cumulative premium development to loss development ratio as written in the question. Because of the perceived ambiguity in the question wording, a scoring adjustment was made for candidates that ignored the word cumulative and mistakenly stopped calculations at PDLD1. This adjustment ensured this perceived ambiguity would not cause a candidate to fail that would have otherwise passed; note filing grade reports and scores may not reflect this grading adjustment.

Part b

The candidates were expected to know one additional method. Most candidates responded with either the empirical method or the Fitzgibbon approach.

Common mistakes included:

- The Feldblum adjustment (removing the fixed portion) as that is not a separate method
- Developing premium directly without any mention to loss

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Part c
Candidates were expected to give one advantage and one disadvantage of the formula method.
A common mistake was switching the advantage and disadvantage, or being unclear as to which was the advantage or disadvantage.